



NATIONAL UNIVERSITY OF TECHNOLOGY

Course : Machine Learning

Department: Artificial Intelligence(2024)-B

Assignment#1

Submitted to: Ma'am Faiza

Group Members

1. [Muhammad Ijlal Asif-F24607072]

2. [Syed Maaz Shah-F24607091]

NUTECH ID: F24607072

SEMESTER: SPRING 2026

Project Proposal: VisionGuard

1. Project Abstract & Introduction

Glaucoma is a leading cause of permanent blindness worldwide. Because it has no early symptoms, finding it before severe damage happens is critical. Currently, doctors must manually check complex eye scans, which takes a lot of time and requires special training. **VisionGuard** is an automated AI system designed to solve this problem using a **Dual-Stream Convolutional Neural Network (CNN)**. Most AI models only look at one type of image. To make our project stand out, we are using a **Multi-Modal** approach. Our model will be trained to look at two different things at the same time: **Retinal Fundus Photos** (2D color pictures of the eye) and **OCT Scans** (3D structural scans). By combining information from both types of scans, our AI will be much more accurate than models that only use one type. **Motivation:**

- **Real-World Impact:** Creating a reliable tool that helps catch glaucoma early, preventing blindness.
- **Saving Time:** Automating the screening process so doctors can focus on patients who need surgery immediately.
- **Higher Accuracy:** Using two types of scans together (Multi-Modal) gives the AI a complete picture, leading to better results.

2. Overall Aims & Achievable Targets

Our main goal is to build a highly accurate Machine Learning model that can classify eye scans as healthy or glaucomatous. **Achievable Targets:**

- **Data Pre-processing:** Clean and resize images so the AI can learn from them easily.
- **Model Development:** Build a Dual-Stream CNN that can extract patterns from

both the 2D color images and the 3D scans, and then combine those patterns.

- **Explainable AI (XAI):** We will use a technique called **Grad-CAM**. This will generate "heatmaps" on the eye images to show the doctor exactly *where* the AI found signs of glaucoma. This builds trust in the AI.

3. Dataset & Performance Metrics

Dataset: We will use the **REFUGE (Retinal Fundus Glaucoma Challenge)** or **ACRIMA** dataset. These are free, high-quality medical datasets containing hundreds of eye scans that have already been correctly labeled by professional eye doctors.

Performance Metrics: We will evaluate our model using standard, professional metrics:

- **Accuracy:** The overall percentage of correct predictions.
- **Sensitivity (Recall):** How well the AI catches actual positive cases of glaucoma (very important in healthcare).
- **Specificity:** How well the AI avoids giving false alarms to healthy patients.
- **F1-Score:** To ensure the model performs well overall, balancing both positive and negative predictions.

4. Potential Application Areas

- **Rural Clinics:** Our AI can act as a screening tool in areas where there are no eye specialists available.
- **Telemedicine:** Patients can get their eyes scanned locally, and the AI can automatically flag urgent cases for remote doctors to review.
- **Second Opinion Tool:** Assisting busy doctors in large hospitals by double-checking edge cases to reduce human error.